

Examples of Mixtures



MIXTURE

teachoo

Dissolving of sugar in water



↑
Sugar



↑
Sugar Solution

NAME: _____

YEAR: _____

KWA-THEMA SKILL SCHOOL
LA: NS TECH SIMPLIFIED ACTIVITIES
YEAR 3 TERM 2
2024

CONTENT OVER VIEWTERM2: MATTER AND MATERIALS

MATTER AND MATERIALS	STATE OF MATTER Solid, liquid and gases • Mix and match • State which column matches each state • States which properties is best described by the ff term	MIXTURES • Which statement best describe the ff concepts • Which mixture is the pure substance amongst the following? • How to separate mixtures	ACID AND BASES • Choose the best definition • What is the difference between Acid and bases? • Sort the various substances by using litmus paper as acid, bases, neutrals • Practical activity
	METALS AND NON-METALS • Match the ff concepts with the correct description • State whether the ff are non-metals or metals • How to use metals and non -metals	SOLUTIONS AS SPECIAL MIXTURES • Ingredients of a mixture • Explain the difference between a mixture and solution • Which one best describe	PROCESS TO PURIFYWATER Clean Water • Introduce and explain concepts with example • Importance of drinking clean water • Method of how to

		properties of mixture	clean water
		Dissolving • Concept explanation • Rate of dissolving Practical investigation: Rate of dissolving	<u>Impact of weather, temperature and wind on matter and materials.</u> • Impact of weather on materials Ways to protect materials
	Processing material • Best description of the concept (matching concepts with the description) • How to make new material or products (follow all processes) • List the properties of raw material used to make new products • Practical demonstration the changes of raw material	The Water cycle • Diagram to explain different concepts • Name the stages of the water cycle • Draw the water cycle and match each description with concepts	

KWA-THEMA SKILL SCHOOL

SIMPLIFIED ACTIVITIES

NS -TECH YEAR 3 TERM 2

2024

NAME: _____ **DATE:** _____

STAND: MATTER AND MATERIALS

TOPIC: MIXTURE

In this topic we are going to look at mixtures, their properties, method of separation and distinguish between a mixture and pure substance.

1.1 DEFINATION OF CONCEPTS/STATEMENTS AND DISCRIPTION(BEST)

1.SIMP: STUDY THE FOLLOWING CONCEPTS ON YOUR WORK BOOK THEN ANSWER THE FOLLOWING QUESTIONS

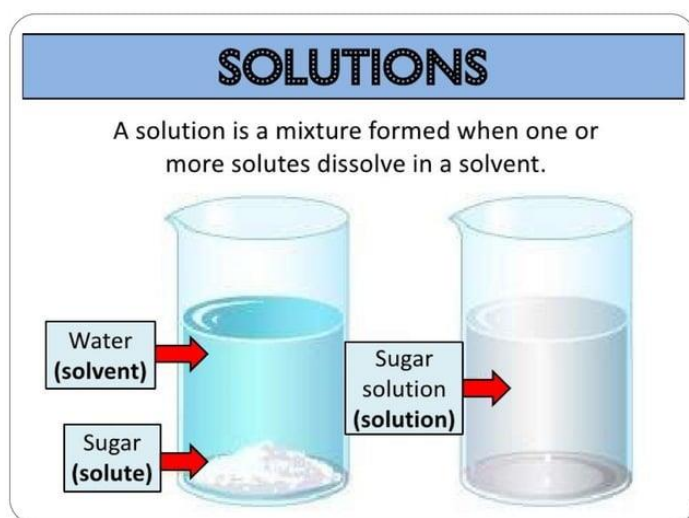
1.SOLUTION

2.MIXTURE

3.SOLID

4. SOLUBLE SUBSTANCE

5. INSOLUBLE SUBSTANCE



Mixtures

A mixture is anything made by combining two or more things.

Some mixtures can be separated and some can't.



What is a **mixture**?

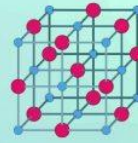
- A **mixture** consists of at least **two different materials mixed together**.
- In some mixtures the different substances are still clearly visible after mixing.
- In others especially mixing of liquids it is difficult to see the original materials.



What Is a Solid?



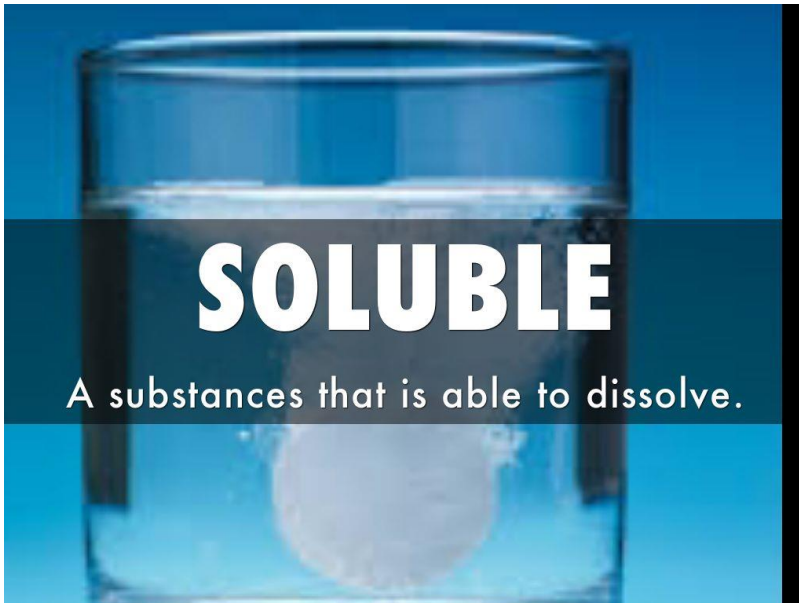
- A solid is a state of matter with fixed shape and volume.
- Particles are close-packed so they can vibrate, but not flow.
- Solids are rigid.
- They cannot be compressed easily.



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SOLUBLE

A substances that is able to dissolve.



Soluble/Insoluble

A soluble substance can be dissolved.

If something will not dissolve, it is ***insoluble***. If two or more chemicals are mixed in a solution and a solid appears, the solid is insoluble. In this case, the solid is called a ***precipitate***.

Experiment of Soluble and Insoluble Substance



Soluble vs. Insoluble



1.LOOK AT THE ABOVE BOX THERE ARE DIFFERENT CONCEPTS THEN MATCH THEM WITH THE CORRECT DISCRPTION BELOW

DESCRIPTION STATEMENTS	FIT IN MATCHING CONCEPTS
1.One of the 3 state of matter has a fixed shape. Solid can be soluble or insoluble	
1.2 Mixture is two or more materials that have been mixed together and, in some mixture, the different materials are still clearly visible after mixing	
1.3 This is a special kind of mixture where components are uniform	
1.4 These substances they form solution when they are mixed with water	
1.5 These substances they do not form solutions when they are mixed with water	

(5)

ACTIVITY 2

1.WHAT IS THE MIXTURE?

(2)

2. Complete the following sentence by writing SOLUBLE and INSOLUBLE in the open space

2.1 Substance that DO NOT form solution when they are mixed with water are called _____ substance

2.2 Substance that forms solutions when they are mixed with water are called _____ substances

2.3 Give an example of a pure substance _____

(3)

MIXTURES

Mixtures Definition - A mixture contains two or more elements or compounds that are not chemically combined together.

PURE SUBSTANCE

- Pure substances are mostly homogeneous in nature containing only one type of atom or molecule.
- These substances mainly have a constant or uniform composition throughout.
- The substances have fixed boiling and melting points.
- A pure substance usually participates in a chemical reaction to form predictable products.



FRUIT SALAD



PAELLA



MILK AND CEREALS



OIL AND WATER

EXAMPLES OF PURE SUBSTANCE



Pure Substance OR Mixture

- Water



- Pizza



- Sand



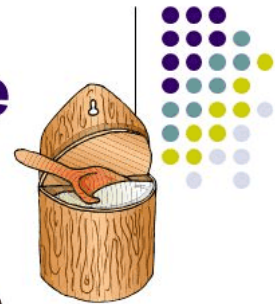
- Honey



- Cereal



- Sugar



- Flour



- Laundry detergent (contains white and blue crystals,



- Jello



Pure Substances	Mixtures
compounds and elements	homogeneous and heterogeneous mixtures
made of molecules or atoms that are all the same in a sample	made of molecules, atoms, or a combination that are different in the sample
definite composition	no definite composition
single chemical formula	no single chemical formula
chemically made and chemically separated (compounds only)	physically mixed and can be physically separated as particles retain their unique properties

Pure Substances vs. Mixtures

Pure Substances

1. All samples have the same physical and chemical properties.
2. Constant composition = All samples have the same pieces in the same percentages.
3. Homogeneous.
4. Separate into components based on **chemical properties**.
5. Temperature stays constant while melting or boiling.

Mixtures

1. Different samples may show different properties.
2. Variable composition = Samples made with the same pure substances may have different percentages.
3. Homogeneous or heterogeneous.
4. Separate into components based on **physical properties**.
5. Temperature usually changes while melting or boiling because composition changes.

1.2 COMPILE OWN DIFFERENCE BETWEEN PURE SUBSTANCE AND MIXTURE USING ALL INFORMATION GATHERED FROM YOUR PREVIOUS LESSONS

PURE SUBSTANCES	MIXTURES
1.	1.
2.	2.
3.	3.

(6)

**HOW TO SEPARATE MIXTURES
DIFFERENT METHODS**

- Handpicking
- Threshing
- Winnowing
- Sieving
- Evaporation
- Distillation
- Filtration or Sedimentation

Separating Mixtures - Winnowing



SEPARATING MIXTURES

SIEVING

Can separate large particles from small particles.

SIEVE

- The sand and rock mixture is shaken through.
- The small holes in the sieve let the sand particles through and prevent the larger rocks from passing through.



FILTRATION

Can separate solids that are insoluble from a liquid.

FILTER PAPER

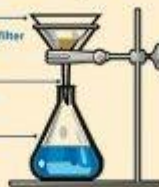
Sand gathers in the filter paper. Water passes through.

FUNNEL

Channels the water into the flask.

FLASK

Gathers the water after filtration.



EVAPORATION

Can separate solids that are soluble from a liquid.



SALT WATER

EVAPORATING DISH

Water evaporates into the air, leaving only the salt.

BUNSEN BURNER

Heats the salt water so that it boils and turns to steam.

CONDENSATION

Condensing is when water vapour changes into liquid water.



COLD SURFACE

The steam hits the surface and condenses into water droplets.

BUNSEN BURNER

Heats the water so that it boils and turns to steam.

CHROMATOGRAPHY

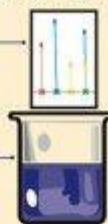
Can separate different colour dyes.

PAPER

- Spots of ink are placed on a pencil line.
- Solvent soaks up to the ink and each colour keeps upwards at a different rate.

SOLVENT

The paper is placed in the solvent which soaks up to the ink spots.



DISTILLATION

Can separate a solvent from a solution.

FLASK

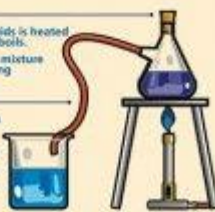
- A mixture of liquids is heated from below and boils.
- The water in the mixture evaporates leaving the remainder.

CONDENSER

Steam condenses in the tube.

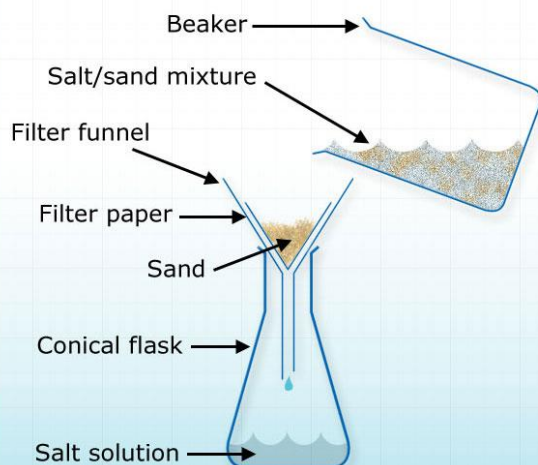
BEAKER

Pure, distilled water collects in the beaker.



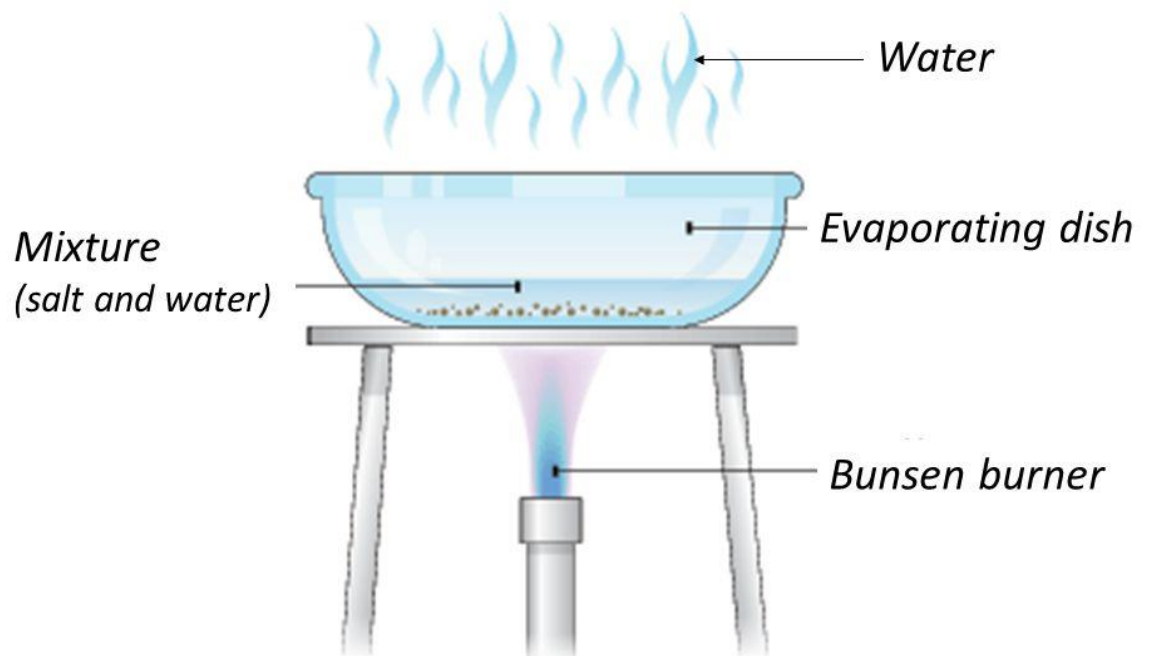
Sand and Salt Separation Practical

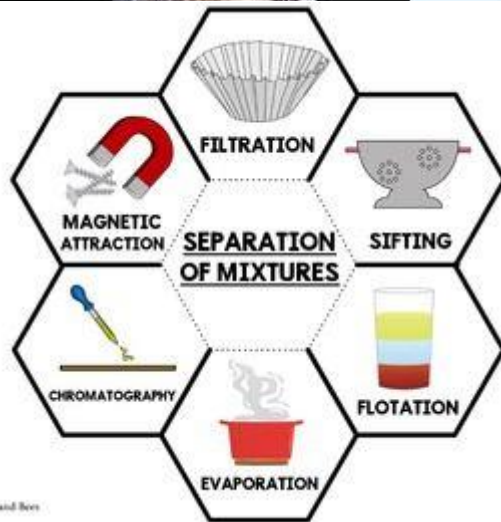
Filtering the salt-sand-water mixture



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Separating Mixtures: **Evaporation**





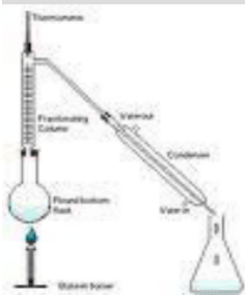
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4 ways to separate a mixture

Can be separated by PHYSICAL
means, do NOT need chemical
ways to separate mixtures



Separating Mixtures

We have talked about four methods of separation. Write the name under the correct picture and give a short description of what how it works.

	#1. 	#2. 	#3. 	#4. 
Name:				
Description:				

Use your understanding of the above techniques to answer the questions below:

5. I have a mixture of salt, water, and iron. I pass them through a filter. What is left in the filter and what passes through into the cup?	6. I have a mixture of tomato sauce, sugar, and carrots. I bring a magnet close by. What does the magnet pick up and what is left behind?	7. I have a mixture of rocks, sand, and water. I pass it through a screen. What is left on top of the screen and what has fallen through the screen?
8. I have a mixture of iron, salt, and water. I heat it up to a boil. Explain what happens to each part of my mixture.	9. I have a mixture of iron, salt, and water. I bring a magnet close by. What does the magnet pick up and what is left behind?	10. I have a mixture of iron, salt, and water. I pass them through a screen. What stays on top and what passes through?

PROPERTIES OF MIXTURE

- The components retain their original physical and chemical properties without any changes.
- The components do not share any chemical bonding while in a mixture.
- The mixture can be separated into its components physically by using some techniques.
- The proportions of the components are variable and can vary in quantity.

Learn more:

ACTIVITY

MULTIPLE QUESTIONS CIRCLE THE CORRECT ANSWER

1 SOLIDS

a. Can be soluble or insoluble

b. Water is one of the solid material

2.Which of the following best describes chemical mixture

a. When two substance are combined but are not chemically bound

b. Substance made through chemical bonding

3.What type of substance is water

a. Colloid

pure substance

4.Which statement is true about the properties of mixtures

a. Sometimes a mixture can be separated by physical process

b. A mixture has no composition of any different material (4)

THE FF DRAWING

1.Idicate/say what is the content in each container

Solution



Water + Salt

Mixture



Water + Sand

Name: _____

Date: _____

Mixture and Solutions

D A S B P N E A X B X O A H Q O D
G I P J Z B V Y G V A K J N I K W
Z T H A P K L X U N C Q R C V A O
S A R N O G O Y A L C U E R T D U
U E G H U U S P D G M C Z E T L O
B G D D V E S C A I R L R D Y D P
S D Q P H U I A X E A X V L G V I
T K V N E F D T A D G S W N D J H
A H M J G D U M K O H K A X K E Z
N J H F I R W A T J S T E L S T U
C E D Q E M E J K K B G H R T A C
E A N J Y L E M O N A D E I T R A
S L A P U P K S F C R Q N P N A Z
T Y S M Q U S E F V R H G S V P E
I I P G Q I J P V S U B O W O E K
M J S O L U T I O N C M P Q T S T
V L L H G I S O L V E N T V A J A

dissolve

mixture

separate

substances

icecream

salt

solution

water

lemonade

sand

solvent

SOLUTION AS SPECIAL MIXTURE

INGREDIENTS OF A MIXTURE PLUS PRACTICAL DEMONSTRATION

FOR (15) GIVE YOUR OWN EXAMPLE IN A BOX BELOW

INGREDIENTS	HOW YOU MIX IT
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.

COMPLETE PROCESS /PICTURE OF YOUR PROJECT

--

MIXTURE CONSIST

Definition and Examples

- A mixture consists of two or more chemically distinct components that do not react with each other.
- Mixtures can be solids, liquids, gases, or a combination of states of matter.
- A homogeneous mixture has a uniform composition, while a heterogeneous mixture has a non-uniform composition.
- Examples of mixtures include air, salt water, brass, rocks, and soft drinks

SOLUTION CONSIST OF

A chemical solution exhibits several properties:

- A solution consists of a homogeneous mixture.
- A solution is composed of one phase (e.g., solid, liquid, gas).
- Particles in a solution are not visible to the naked eye.
- A solution does not scatter a light beam.
- Components of a solution cannot be separated using simple mechanical filtration.

SOLUTIONS ARE SPECIAL KIND OF MIXTURES

ACTIVITY

WHEN IS A MIXTURE A SOLUTION

WE USE THESE EXAMPLES

- **We mix substance with water to make solution**
- **Sugar, salt, sand, oil, vinegar, flour, copper sulphate**
- **Tap water**
- **glass container**
- **Plastic spoon to scoop and stirring**

Instruction

- **Half fill glass cup with water**
- **A scoop of sugar (in a glass of water, stir well**
- **Look at the mixture and discuss what it looks like**

RECORD YOUR OBSERVAT

SUBSTANCE	OBSERVATIONS	
	NON-OF THE SUBSTANCE HAS DISOLVE	ALL/MOST OF THE SUBSTANCE HAS DISSOLVED
SUGAR		
SALT		
SAND		
OIL		
VINEGAR		
COPPER SULPHATE		

(12)

WHAT IS A SOLUTION

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1.



A solution is a homogenous mixture of two or more substances¹²³.

- The substances can be varied continuously up to what is called the limit of solubility¹.
- A solution may exist in any phase².
- A solution consists of a solute and a solvent².
- The solute is the substance that is dissolved in the solvent².
- The amount of solute that can be dissolved in solvent is called its solubility².
- The solvent is the material that is present in the maximum amount, whereas the solute is the substance that is present in the least amount³.

L

DEFINATIPON OF SOLUTION

A solution is a homogeneous mixture of two or more substances. A solution may exist in any phase.

A solution consists of a solute and a solvent. The solute is the substance that is dissolved in the solvent. The amount of solute that can be dissolved in solvent is called its solubility. For example, in a saline solution, salt is the solute dissolved in water as the solvent.

For solutions with components in the same phase, the substances present in lower concentration are solutes, while the substance present in highest abundance is the solvent. Using air as an example, oxygen and carbon dioxide gases are solutes, while nitrogen gas is the solvent.

Note that whether or not components start in different phases, a solution only consists of one phase. For example, sugar is a solid, while water is a liquid, However, a solution of sugar water is liquid only.

FROM THE ABOVE ACTIVITY

1. Which one was the SOLUTE

2.Which one was the SOLVENT

3.Which one is the SOLUTION

(6)

ACTIVITY: WHICH MIXTURES ARE SOLUTIONS?

(use previous activity to answer)

4.When is the mixture a solution

2

5.What do we call substance that dissolves in water ?

(1)

6.What do we call water in the mixture ?

1

7.What name do we give those mixtures ?

2

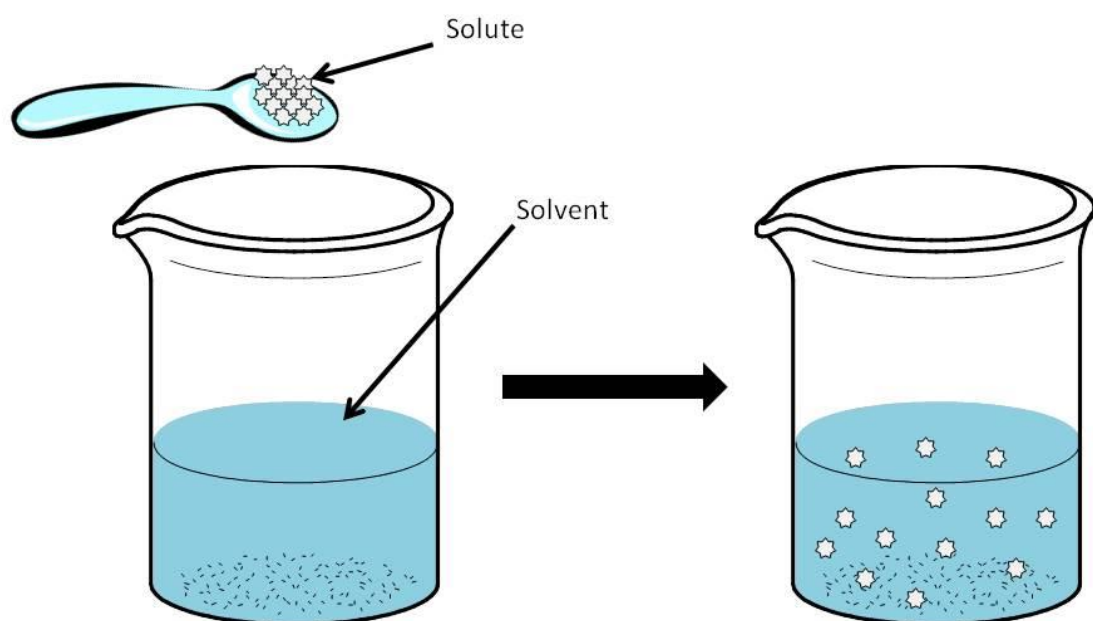
COMPLETE THE FF TABLE AND INDICATE WETHER THE MIXTURE IS SOLUTION(YES) OR (NOT)

MIXTURE	Is the mixture a solution after stirring YES OR NO
Sugar and Water	
Salt and Water	
Sand and Water	
Oil and Water	
Vinegar and Water	

(6)

DISSOLVING

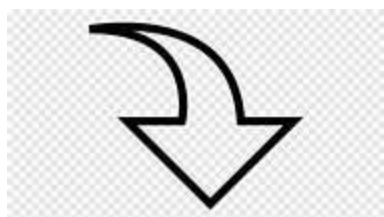
WHAT IS DISSOLVING?

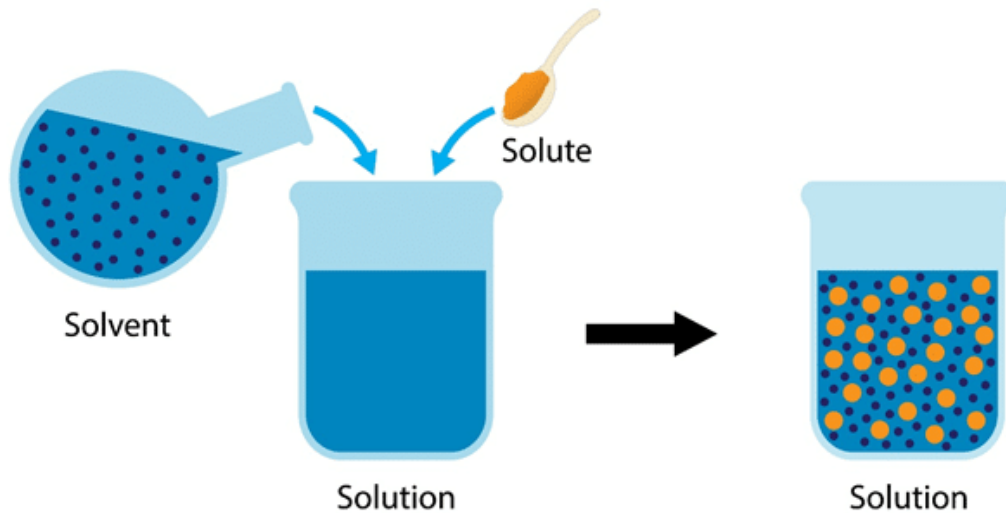


The **solute** is the substance that gets dissolved, the **solvent** is the substance that does the dissolving.

Once mixed, the solute will dissolve and become evenly dispersed within the solvent.

EXPLAINING DISSOLVING CONCEPTS





Dissolving involve two materials separating into parts and the resulting solution is the mixture of the two material.

Example: sugar dissolve in water, salt dissolve in water RATE OF DISSOLVING REFFERS TO: How quickly a solute dissolve in a solvent

Rate of Dissolving

Determined by

- temperature
- stirring
- particle size

PRACTICAL DEMONSTRATION

AIM: INVESTIGATE HOW TEMPERATURE AFFECTS RATES OF DISSOLVING

MATERIAL AND APARATUS		METHOD
3.Clear containers Sugar Hot water Tap water Ice water Tea spoon Clock/watch with second		Fill in the 3 containers with, hot, tap water, ice water Put 5ml teaspoon of sugar in hot water (stir teaspoon) measure how slow/fast the salt dissolve in water Put 5ml teaspoon of sugar in tap water (stir teaspoon) measure how slow/fast the salt dissolve in water Put 5ml teaspoon of sugar in ice water (stir teaspoon) measure how slow/fast the salt dissolve in water
RESULTS		
situation	Time it takes to dissolve	
Sugar in hot		
Sugar in tap		
Sugar in ice		
(3)		

1.What was the aim of the experiment?

_____ **2**

2.Name the 3 things determining the rate of dissolving?

_____, _____, _____ **3**

3.We measure the temperature of the water, what else did we measure_____ **1**

4.Complete by filling missing words.

The sugar dissolved _____ in hot water than the cold water

5.Temperature affects rates of dissolving when we increase temperature of the solvent

PRACTICAL ACTIVITY =(30) OWN INVESTIGATION

AIM:INVESTIGATE HOW GRAIN SIZE AFFECT THE RATE OF DISSOLVING USE

1.FINE SALT

2.CORSE SALT

3.Write own material starting with containers

MATERIAL AND APARATUS		METHOD
1.__. clear containers 2. 3. 4. 5.		Fill in the 3 containers with, _____, _____, _____ Put _____ of _____ in _____ water (stir teaspoon) measure how slow/fast the _____ dissolve in water Put _____ of _____ in _____ water (stir teaspoon) measure how slow/fast the _____ dissolve in water
RESULTS		
situation	Time it takes to dissolve	
in		
in		
		22

DRAW OWN DIAGRAM OF THE ABOVE

EXPANDED/extra activity/practice/addition for those behind

1.What was the aim of experiment

_____ **1**

2.What did we measure?

_____ **2**

3The fine salts dissolve _____ than the coarse salt 1

4.Does the grain size affect the rate of dissolving explain

_____ **2**

PRACTICAL :3

INVESTIGATE HOW STIRRING AFFECTS DISSOLVING

YOUR MATERIAL SHOULD BE (2clear containers, fine salt, tab water, teaspoon

2.Design own method

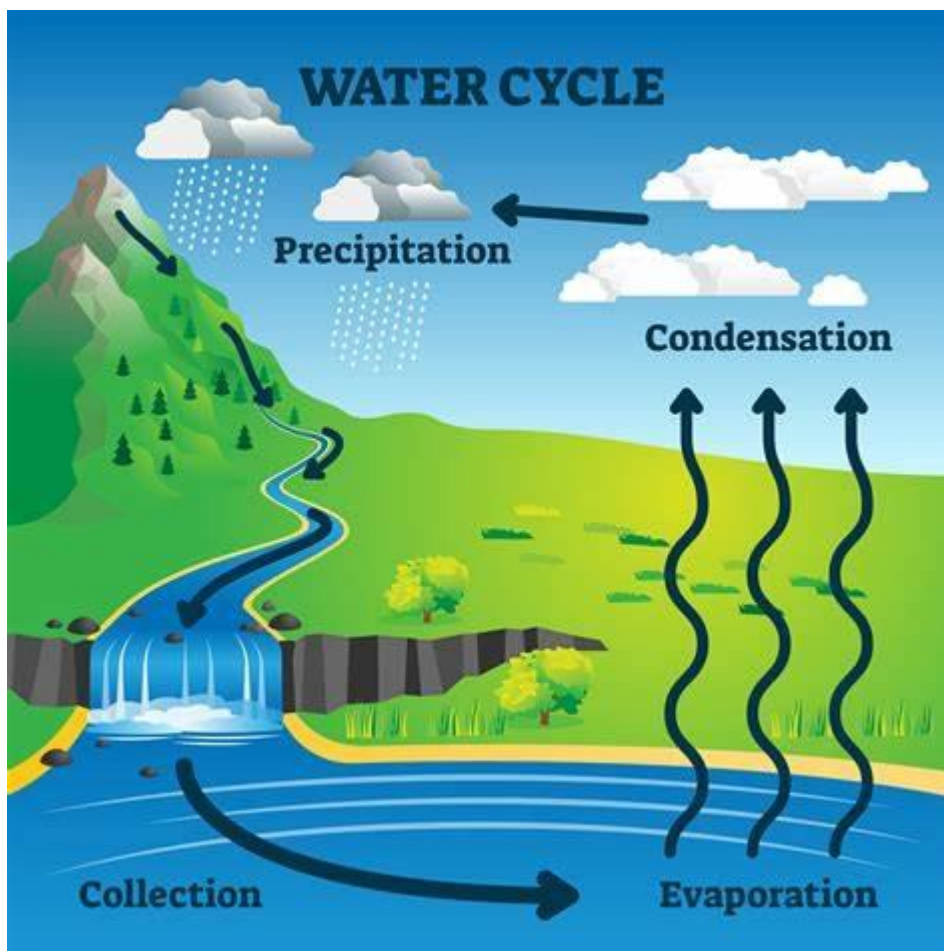
3.Results

(15)

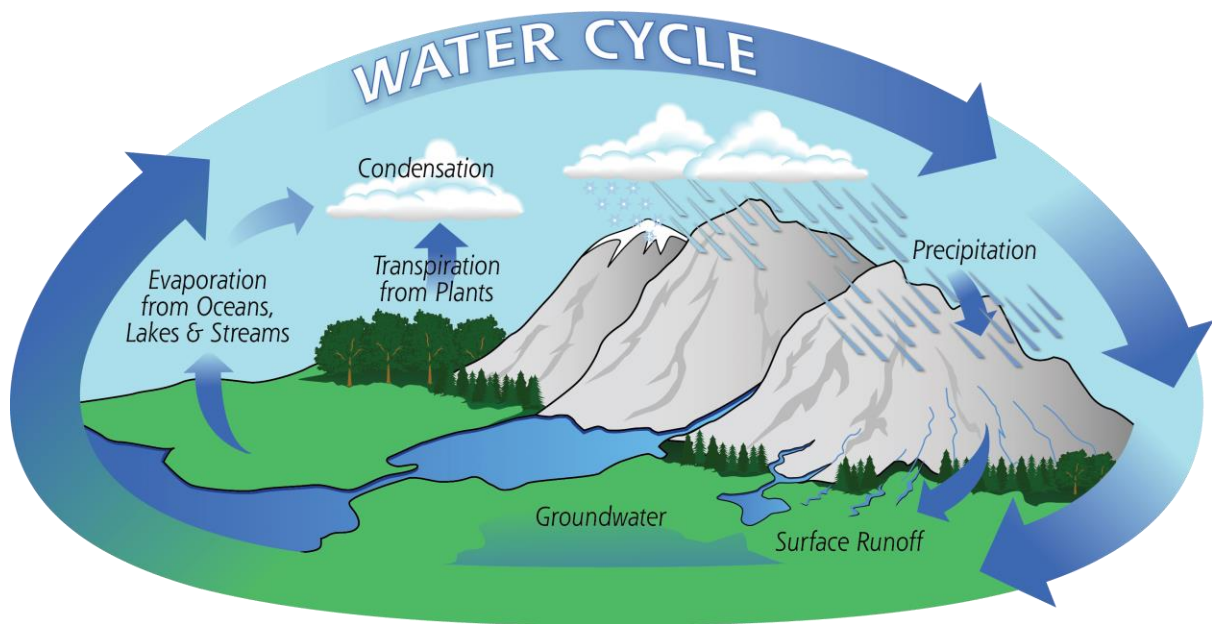
MATERIAL AND APARATUS	METHOD
1. ____ . Clear containers 2. 3. 4. 5.	Fill in the ____ containers with, _____, _____, _____ Put _____ of _____ in
RESULTS	

situation	Time it takes to dissolve	
in		
in		

THE WATER CYCLE



STAGES OF THE WATER CYCLE



Stages of Water Cycle

- 1. Evaporation The sun is the ultimate source of energy, and it powers most of the evaporation that occurs on earth. ...
- 2. Sublimation Sublimation occurs when snow or ice changes directly into water vapour without becoming water. ...
- 3. Condensation The water vapour that accumulated in the atmosphere eventually cools down due to the low temperatures found at high altitudes. ...
- 4. Precipitation ...
- 5. Infiltration ...
- 6. Runoff

MATCH THE DESCRIPTION WITH THE CORRECT CONCEPTS

PRECIPITATION, EVAPOURATION, CONDENSATION, RUN OFF, SUBLIMATION

Discription	Concepts
1. When snow or ice changes to water vapour without becoming water	
2. Water vapour that is accumulated in the atmosphere	
3. The sun is the altimate source of energy	